CART2INSIGHTS: DECODING E-COMMERCE PERFORMANCE

1. PROJECT OVERVIEW

Cart2Insights is an e-commerce data analytics project designed to evaluate sales performance, order fulfilment, product category revenue, payment methods, and customer satisfaction using the O list e-commerce dataset.

The project uses Python, SQL, statistical analysis, and Streamlit to transform raw e-commerce data into meaningful business insights and interactive visualizations.

2. PROBLEM STATEMENT

E-commerce businesses need to understand sales trends, identify high-performing product categories, monitor delivery performance, and evaluate customer satisfaction. This project analyses historical e-commerce data to identify performance patterns and provide data-driven recommendations for improving operational efficiency and customer experience.

3. PROJECT OBJECTIVES

- Analyze order volume and order status distribution.
- Evaluate product revenue and monthly sales trends.
- Identify high-revenue product categories and sellers.
- Analyze payment methods and customer review scores.
- Investigate the relationship between delivery delays and customer satisfaction.
- Develop an interactive dashboard to communicate key findings.

4. DATASET DESCRIPTION

The project uses nine CSV datasets:

- Customers
- Orders
- Geolocation
- Order Items
- Payments
- Reviews
- Products
- Sellers
- Product Category Translation

These datasets contain information about customers, purchases, products, sellers, payments, delivery dates, and customer feedback.

5. TOOLS AND TECHNOLOGIES

- **Python:** Data processing and exploratory data analysis.
- **Pandas:** Data cleaning, transformation, and aggregation.
- **Matplotlib:** Exploratory data visualizations.
- **SQL / MySQL:** Business queries and KPI calculations.
- **Statistical analysis:** Hypothesis testing and relationship analysis.
- **Streamlit:** Interactive dashboard development.

6. METHODOLOGY

a. Loaded and inspected the nine datasets.

b. Examined missing values and data quality issues.

c. Performed exploratory data analysis using Python.

d. Executed SQL queries to analyze orders, revenue, payments, categories, sellers, and delivery performance.

e. Conducted statistical tests to investigate differences in review scores and associations between selected variables.

f. Developed a Streamlit dashboard containing KPI cards, charts, filters, and business recommendations.

7. KEY FINDINGS

METRIC	RESULT
Total orders	99,441
Delivered orders	96,478
Product item revenue	13,591,643.70
Late-delivery rate	8.11%
Repeat-customer rate	3.12%
Average customer review score	4.09 / 5
Reviews with 4–5 stars	77.1%

Health & Beauty was the highest-revenue product category in the analysis.

The average review score was 4.29 out of 5 for on-time deliveries and 2.57 out of 5 for late deliveries, a difference of 1.72 points.

8. BUSINESS RECOMMENDATIONS

Delivery Performance

Investigate delayed deliveries, assess seller dispatch processes, and improve delivery tracking and customer communication.

Customer Retention

Investigate repeat-purchase patterns and evaluate customer engagement or loyalty initiatives.

Product Category Performance

Monitor demand, stock availability, profit margins, and promotional performance across product categories before allocating additional resources.

Customer Satisfaction

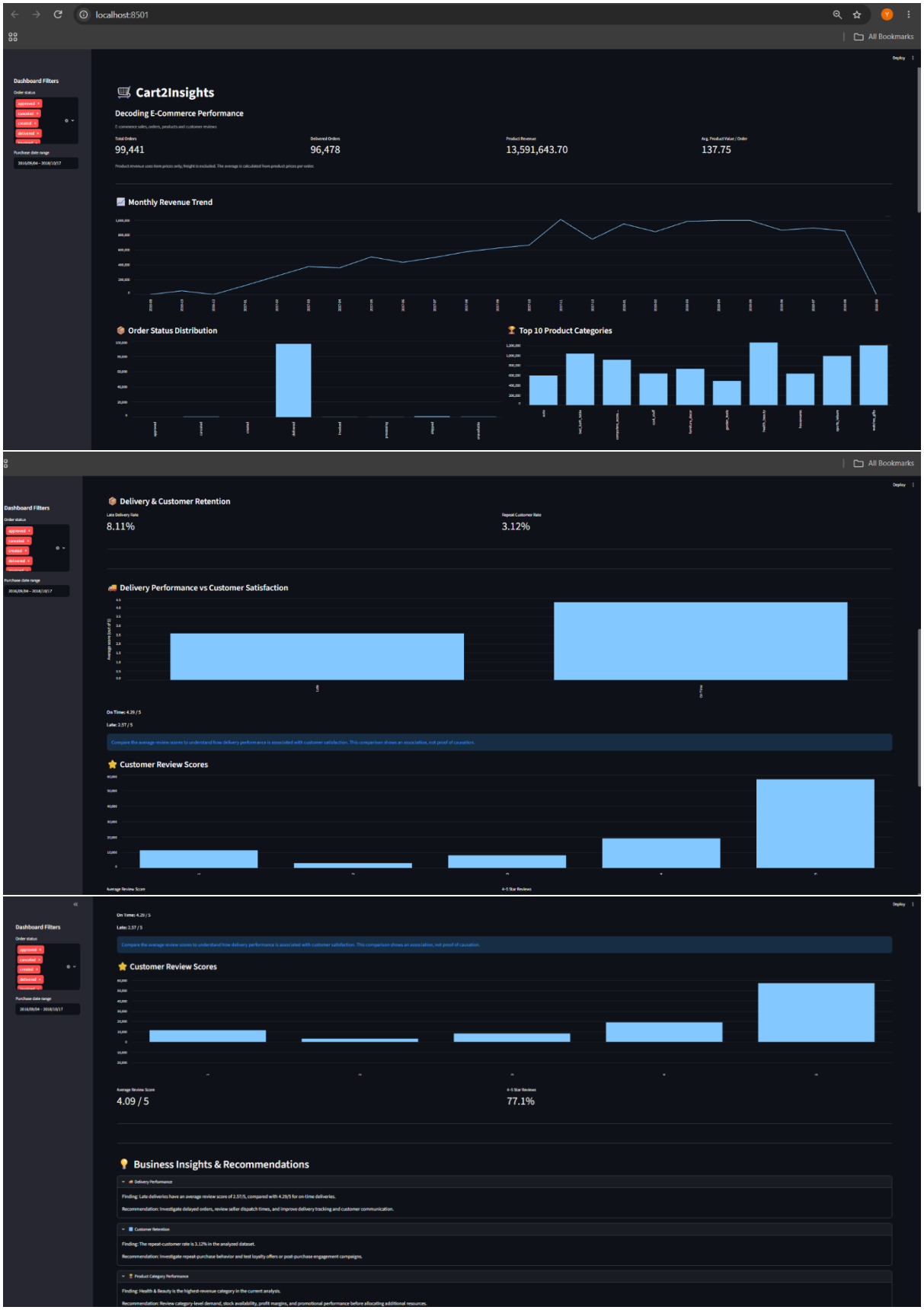
Track delivery-related review scores over time to identify operational issues associated with lower customer ratings.

9. DASHBOARD OVERVIEW

The Streamlit dashboard includes:

- Total and delivered order KPIs.
- Product revenue and average product value per order.
- Monthly revenue trend.
- Order status distribution.
- Top ten product categories by revenue.
- Customer review score distribution.
- Delivery performance versus customer satisfaction.
- Business insights and recommendations.

SCREEN SHOTS OF OUTPUT



10. CONCLUSION

- Cart2Insights demonstrates how Python, SQL, statistical analysis, and interactive visualization can be combined to analyze e-commerce performance. The findings highlight delivery performance, customer retention, and product category revenue as important areas for further investigation.
- The dashboard provides a consolidated view of key metrics and supports data-driven discussions about operational improvement and customer experience.

11. Limitations

- ✓ The analysis is based on historical data. The repeat-customer rate and delivery metrics depend on the definitions and records used in the calculations. Product item revenue is calculated from item prices and excludes freight charges; it should not be interpreted as profit or total customer payments.
- ✓ The observed relationship between late delivery and lower review scores does not establish causation.